CITY MANAGEMENT

Request For Information

Unit and Tenant Information		
To: Tenant Fit Out	RFI#	HMG-KAFD-RFI-ELE-010
	Date	01/03/2026
Tenant.fitout@kafd.sa	Tenant name	HMG
	Building	2.12
	and Unit #	Retail 1, 2,3,4 & 5

RFI Description (Voltage Surge)

With reference to the regulations of the Saudi Electricity Regulatory Authority (SERA) and the attached document titled "Power Quality – SERA Regulatory Framework"

Power Quality

Please be informed that the standard nominal voltage in the Kingdom of Saudi Arabia is 230/400V (Single-phase 230V, Three-phase 400V).

As per SERA regulations, the permissible operating voltage range shall be as follows:

Single-phase: 218.5V to 241.5V

Three-phase: 380V to 420V

However, based on the site measurements and as clearly shown in the attached photographs, the recorded voltage values exceed the allowable limits specified by SERA and go beyond the normal acceptable voltage range in accordance with the above-mentioned regulatory framework.

This voltage deviation represents a non-compliance with SERA standards and may result in serious operational risks, including potential damage to sensitive medical equipment such as portable X-ray machines and other critical devices.

In light of the above, we kindly request your immediate guidance and necessary action to rectify this issue and ensure full compliance with SERA regulations.

Attachments:			
Saudi Electricity Regulating Authority (SERA)	Photograph of measured voltages	Enhancement Drawing - KAFD	

Submitted By:			
Eng. Abdelrahman Kormod	HMG		

Response To RFI (KAFD DMC)			
Response By:	Name, Title	Date	Click or tap to enter a date.
	Company		

6. POWER QUALITY APPLICABLE STANDARDS

6.1. A power quality disturbance exists when there is a deviation from the standards as specified in this section.

6.2. The limits specified in this document for PQ parameters are considered as a compatibility level and shall apply only under normal operating conditions.

6.3. Under normal operating conditions, the compliance criteria will be based on international standards and practices, such as IEC 61000-4-30 and EN 50160. Additionally, SERA will consider PQ field assessments that arise from the implementation of this document, advancements in emerging technologies, evolving policies and regulations, and the continuous development of the electricity sector when establishing and updating these criteria.

6.4. All PQ parameters outlined in the [IEC61000-4-30] standard that are not specifically listed in this section must be measured and recorded. SERA may request this data if deemed necessary.

6.5. Standard for Voltage Variation:

6.5.1. The transmission and distribution voltage levels shall be as defined in SAGC and SADC as shown in Annex 1.

6.5.2. Under normal operating conditions, voltage variations in the transmission and distribution systems, as well as at all connection points shall not exceed ±5% of the nominal voltage, based on SAGC and SADC.

6.5.3. The voltage shall be measured on RMS basis and evaluated for each 10-minute period with measurements undertaken over a period of at least [1 week] based on [IEC 61000-4-30].

SERA - Standard Voltage level

Table 2: Standard Voltage levels and variations defined in the Grid Code

Nominal Voltage	Limits
110 kV	±5%
115 kV	±5%
132 kV	±5%
230 kV	±5%
380 kV	±5%

Table 3: Standard Voltage levels defined in the Distribution Code

Nominal Voltage	Lowest Voltage	Highest Voltage
220 / 127 V	209 / 120V	231 / 134V
380 / 220 V	360 / 209V	400 / 231V
400 / 230 V	380 / 218.5V	420 / 241.5V
13.8kV	13.1kV	14.5kV
33kV	31.4kV	34.7kV
69kV	65.5kV	72.5kV

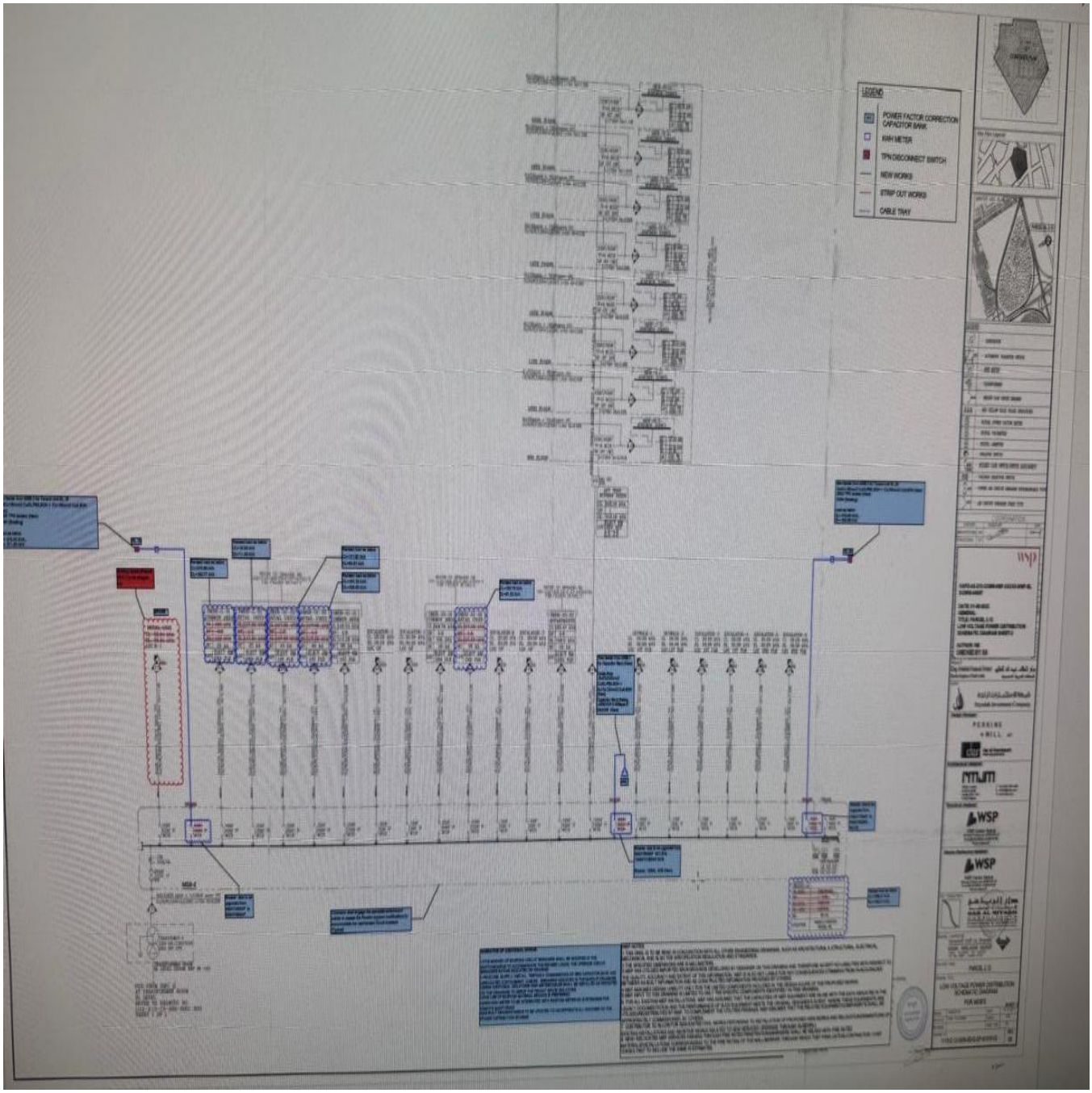
Table 4: Non-standard Voltage levels defined in the Distribution Code

Nominal Voltage	Lowest Voltage	Highest Voltage
11kV	10.45kV	11.55kV
34.5kV	32.78kV	36.23kV

Table 5: Standard for Total Harmonic Distortion

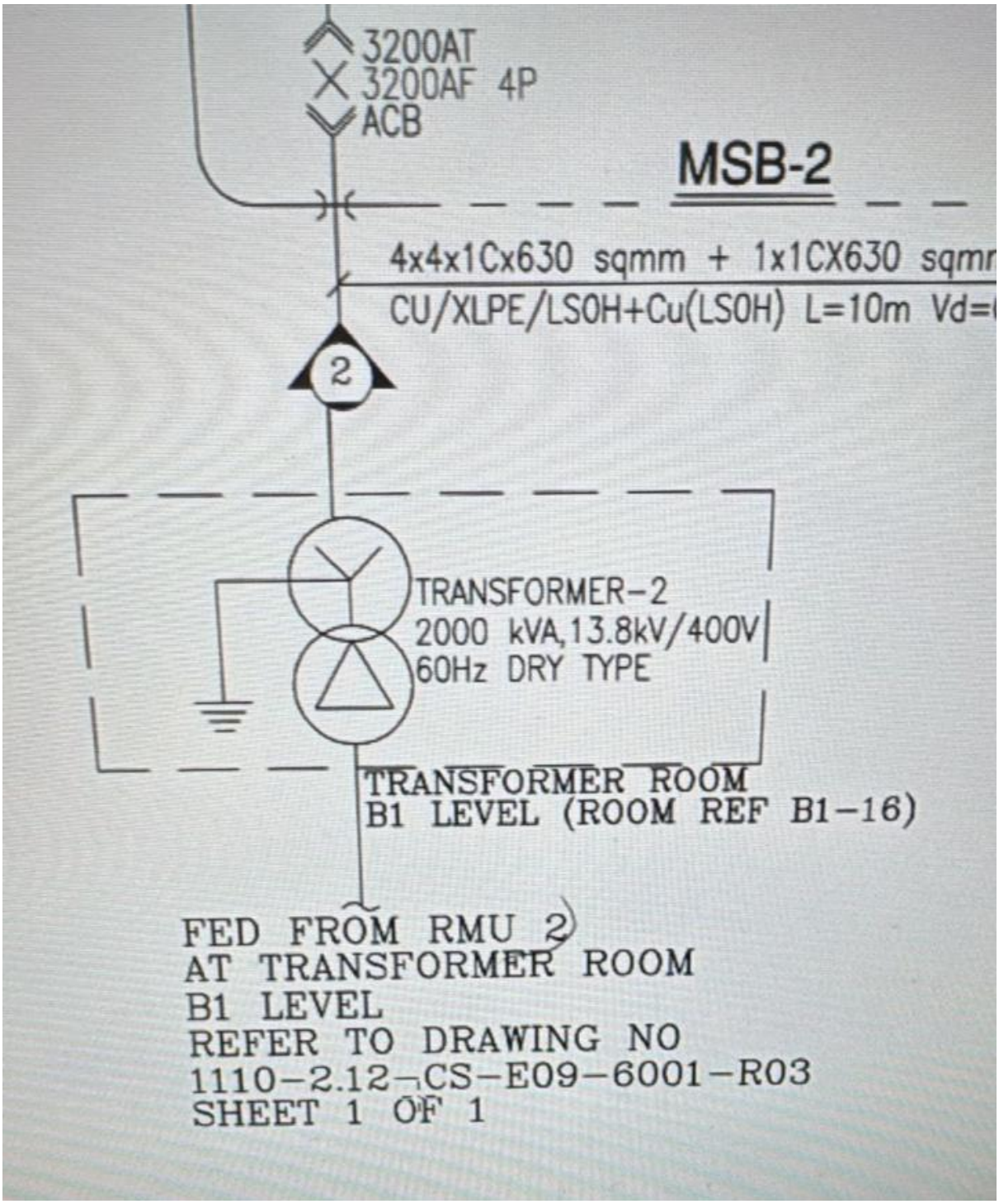
Nominal Voltage	Individual harmonic (%) n ≤ 50	Total Harmonic Distortion THD (%)
380 kV	1.0%	1.5%
230 kV		
132 kV		
115 kV		
110 kV		
33 kV-69 kV	See Tables 5-7	3.0%
11 kV & 13.8 kV		4.0%
230-400 V		5.0%
127-220 V		5.0%

Enhancement Drawing - KAFD 2.12



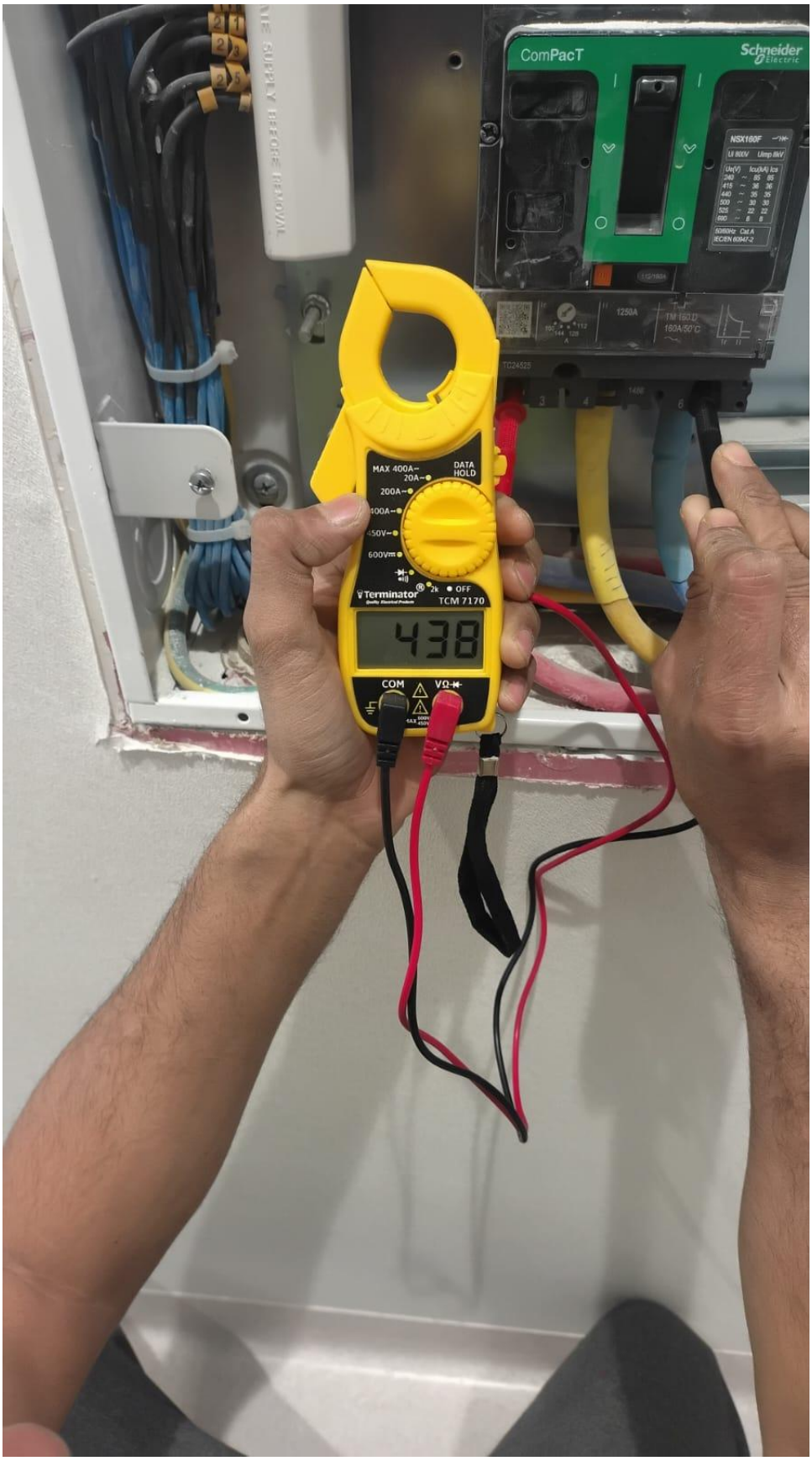
Enhancement Drawing - KAFD 2.12

As per Enhancement drawing received from KAFD clearly mentioned supply voltage is 400V from Transformer-2, rating 2000 kVA, 13.8kV / 400V, 60Hz Dry Type.



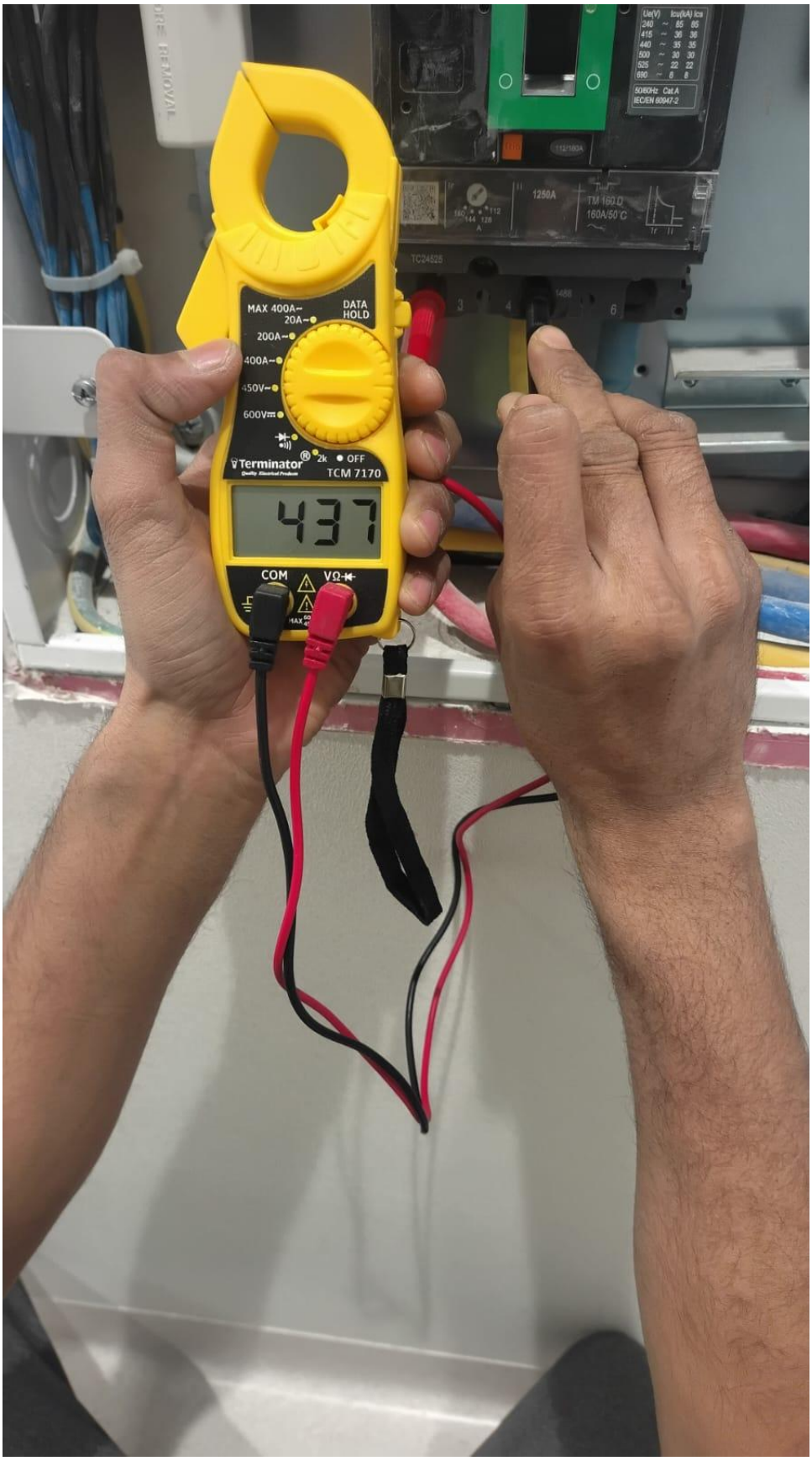
Measured Voltages in KAFD 2.12

Three-phase 438 V



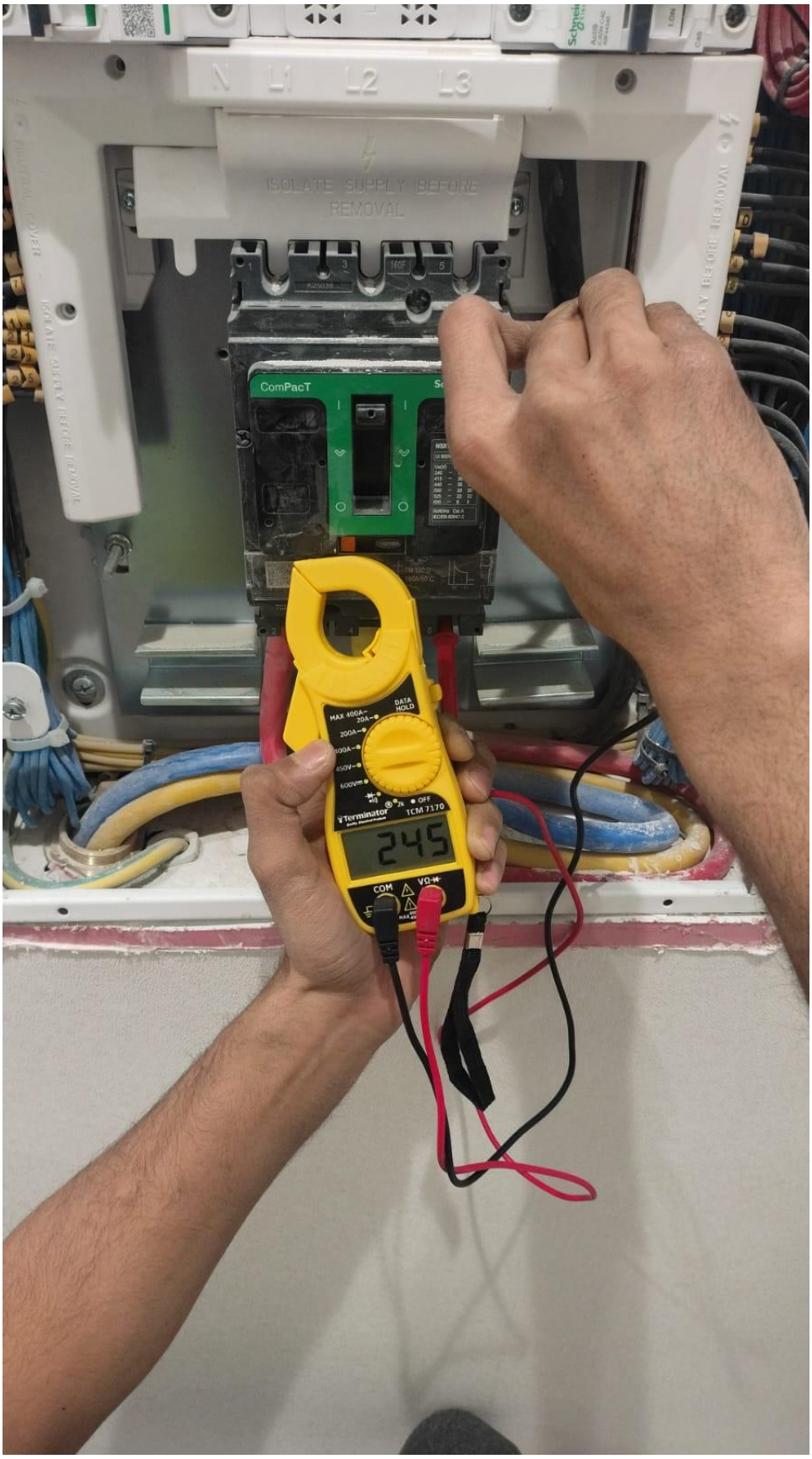
Measured Voltages in KAFD 2.12

Three-phase 437 V



Measured Voltages in KAFD 2.12

Single-phase 246 V

Measured Voltages in KAFD 2.12

Single-phase 245 V